Linear system with LU decomposition :

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Academic Year: 2026 – 2027

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1. What is LU decomposition?

LU decomposition factorizes a square matrix **A** into the product of a **lower triangular** matrix **L** and an **upper triangular** matrix **U**. It is used to solve linear systems **Ax = b** efficiently, to compute determinants and inverses.

A = L · U with L : 1 on the diagonal, zeros above | U : zeros below the diagonal

Why use it?

- Once A = LU is computed, **any** right-hand side b is solved cheaply (two triangular solves).
- Cost of the factorization ≈ $(2/3)n^3$ operations; each extra solve costs only n^2.
- det(A) = product of the diagonal entries of U (when no row swaps are used).

2. Solving Ax = b with LU

Replace A by LU, then split the problem into two easy triangular systems:

Ax = b ⇒ **LUx = b** ⇒ **Ly = b** (forward substitution) then **Ux = y** (backward substitution)

Algorithm (Doolittle method, L has a unit diagonal)

Step	Formula
1. Row i of U (j ≥ i)	$u_{ij} = a_{ij} - \Sigma_{k<i}\, l_{ik} \cdot u_{kj}$
2. Column j of L (i > j)	$l_{ij} = (a_{ij} - \Sigma_{k<j}\, l_{ik} \cdot u_{kj}) / u_{jj}$
3. Forward Ly = b	$y_i = b_i - \Sigma_{k<i}\, l_{ik} \cdot y_k$
4. Backward Ux = y	$x_i = (y_i - \Sigma_{k>i}\, u_{ik} \cdot x_k) / u_{ii}$

Condition: all pivots u_{ii} must be non-zero. Otherwise, use partial pivoting: **PA = LU** (P = permutation matrix).

3. Worked example

Solve the system below using LU decomposition:

$$\begin{aligned} 2x_1 + x_2 + x_3 &= 7 \\ 4x_1 + 3x_2 + 3x_3 &= 19 \\ 8x_1 + 7x_2 + 9x_3 &= 49 \end{aligned}$$

$$\begin{vmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{vmatrix} \cdot \begin{vmatrix} x1 \\ x2 \\ x3 \end{vmatrix} = \begin{vmatrix} 7 \\ 19 \\ 49 \end{vmatrix}$$

Step 1 – Compute L and U

Starting point

$$\begin{vmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{vmatrix}$$

We use Gaussian elimination: **U** is the final upper triangular matrix, and **L** collects the multipliers used along the way. Each multiplier is l = (entry to eliminate) / pivot.

Column 1 (pivot = 2)

- $l_{21} = 4 / 2 = \mathbf{2}$, so $R_2 \leftarrow R_2 - 2 \cdot R_1 = (4–4, 3–2, 3–2) = \mathbf{(0, 1, 1)}$
- $l_{31} = 8 / 2 = \mathbf{4}$, so $R_3 \leftarrow R_3 - 4 \cdot R_1 = (8–8, 7–4, 9–4) = \mathbf{(0, 3, 5)}$

Column 2 (pivot = 1, the new value in position (2,2))

- $l_{32} = 3 / 1 = \mathbf{3}$, so $R_3 \leftarrow R_3 - 3 \cdot R_2 = (0, 3–3, 5–3) = \mathbf{(0, 0, 2)}$

Building U and L

- **U** is the final upper triangular matrix: its rows are $R_1 = (2, 1, 1)$, $R_2 = (0, 1, 1)$ and $R_3 = (0, 0, 2)$.
- **L** has 1s on the diagonal and 0s above it. Each multiplier goes in the position of the entry it eliminated: $l_{21} = 2$, $l_{31} = 4$ and $l_{32} = 3$.

$$\begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{vmatrix} \begin{vmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{vmatrix} = \begin{vmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{vmatrix}$$

L **U** **A**

Check: row 3 of L times column 3 of U = 4·1 + 3·1 + 1·2 = 9 = a_{33} ✓

Step 2 – Forward substitution: solve Ly = b

$$\begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{vmatrix} \begin{vmatrix} y1 \\ y2 \\ y3 \end{vmatrix} = \begin{vmatrix} 7 \\ 19 \\ 49 \end{vmatrix}$$

Unknown	Computation	Value
y1	= 7	7
y2	= 19 – 2·(7)	5
y3	= 49 – 4·(7) – 3·(5)	6

$$\mathbf{y = (7, 5, 6)^T}$$

Step 3 – Backward substitution: solve Ux = y

$$\begin{vmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{vmatrix} \begin{vmatrix} x1 \\ x2 \\ x3 \end{vmatrix} = \begin{vmatrix} 7 \\ 5 \\ 6 \end{vmatrix}$$

Unknown	Computation	Value
x3	= 6 / 2	3
x2	= 5 – 1·(3)	2
x1	= (7 – 1·(2) – 1·(3)) / 2	1

Solution: $x_1 = 1,\quad x_2 = 2,\quad x_3 = 3$

Verification: 2(1)+2+3 = 7 ✓ 4(1)+3(2)+3(3) = 19 ✓ 8(1)+7(2)+9(3) = 49 ✓

Bonus: det(A) = 2 × 1 × 2 = **4** (product of the diagonal of U).

4. Summary

Step	Action
1	Factorize A = LU (find L, U)
2	Solve Ly = b by forward substitution
3	Solve Ux = y by backward substitution

Key points: A = LU • Ly = b then Ux = y • det(A) = $\Pi\, u_{ii}$ • pivoting (PA = LU) if a pivot is zero.